

Service_Module_model_summary.txt

TBD2 pipeline output - rendered to PDF

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.38030394934865536

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9582314936921804

Scenario_9: 0.8038222834527426

Scenario_8: 0.6386992184153006

Scenario_7: 0.4897011323447945

Scenario_6: 0.3610822487815188

Scenario_5: 0.2517311237794721

Scenario_4: 0.15915817873087432

Scenario_3: 0.0901217409995456

Scenario_2: 0.03943721790523324

Scenario_1: 0.011054855384891085

Average MSE across models is 9.763479802480928e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: ACT10126, ACT10127, SCM10111. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: GMD10104, SAN10115, SCP10112. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: SCC10113, SCM10107, SAN10110. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- ACT10126: Compliance Metrics
- ACT10127: Service Activation Compliance Rate
- SCM10111: Capacity Utilization Trends

Scenario_9 (Avg Risk=0.804):

- GMD10104: Real-Time Processing Rate
- SAN10115: Anomaly Escalation Rate
- SCP10112: Customer Feedback on Planned Services

Scenario_8 (Avg Risk=0.639):

- SCC10113: Content Accessibility
- SCM10107: Forecasting and Planning Metrics
- SAN10110: Anomaly Resolution Success Rate

Baseline Predictor (Mean of SRVC_001):

MSE: 0.31679410773333333

R-squared: 0.0

RMSE: 0.5628446568399964

MAPE: 1.4865769739699757%

PyTorch Model Evaluation:

MSE: 0.0003848489084782621

R-squared: 0.9987851765576328

RMSE: 0.01961756632404392

MAPE: 0.04934783518069504%

Model Evaluation Metrics:

MSE: 0.0003848489084782621

RMSE: 0.01961756632404392

R²: 0.9987851765576328

MAPE: 0.04934783518069504%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of Service Module Risk Analysis Summary-----